

Pearson Edexcel Award

Time 3 hours

Paper
reference

9811/01

Advanced Extension Award Mathematics

Insert for questions 5, 6 and 7

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5. An aeroplane leaves a runway and moves with a constant speed of V km/h due north along a straight path inclined at an angle $\arctan\left(\frac{3}{4}\right)$ to the horizontal.

A light aircraft is moving due north in a straight horizontal line in the same vertical plane as the aeroplane, at a height of 3 km above the runway.

The light aircraft is travelling with a constant speed of $2V$ km/h.

At the moment the aeroplane leaves the runway, the light aircraft is at a horizontal distance d km behind the aeroplane.

Both aircraft continue to move with the same trajectories due north.

- (a) Show that the distance, D km, between the two aircraft t hours after the aeroplane leaves the runway satisfies

$$D^2 = \left(\frac{6}{5}Vt - d\right)^2 + \left(\frac{3}{5}Vt - 3\right)^2 \quad (3)$$

Given that the distance between the two aircraft is never less than 2 km,

- (b) find the range of possible values for d . (7)
- (+S1)

(Total for Question 5 is 11 marks)

6.

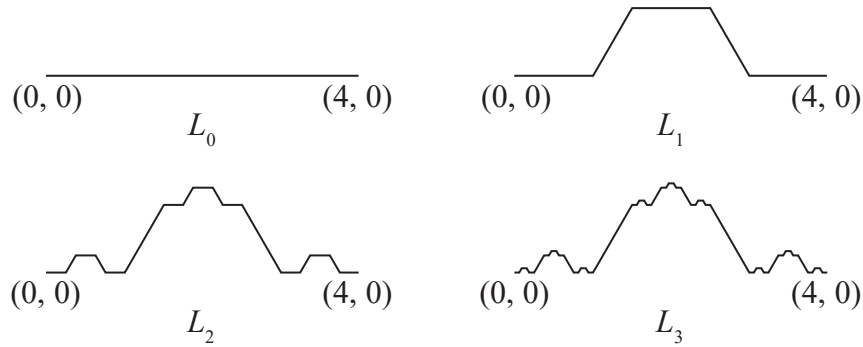


Figure 2

Figure 2 shows the first few iterations in the construction of a curve, L .

Starting with a straight line L_0 of length 4, the middle half of this line is replaced by three sides of a trapezium above L_0 as shown, such that the length of each of these sides is $\frac{1}{4}$ of the length of L_0

After the first iteration each line segment has length one.

In subsequent iterations, each line segment parallel to L_0 similarly has its middle half replaced by three sides of a trapezium above that line segment, with each side $\frac{1}{4}$ the length of that line segment.

Line segments in L_n are either parallel to L_0 or are sloped.

(a) Show that the length of L_2 is $\frac{23}{4}$ (2)

(b) Write down the number of

- (i) line segments in L_n that are parallel to L_0
- (ii) sloped line segments in L_2 that are not in L_1
- (iii) **new** sloped line segments that are created by the $(n + 1)$ th iteration. (3)

(c) Hence find the length of L_n as $n \rightarrow \infty$ (6)

The area enclosed between L_0 and L_n is A_n

(d) Find the value of A_1 (2)

(e) Find, in terms of n , an expression for $A_{n+1} - A_n$ (3)

(f) Hence find the value of A_n as $n \rightarrow \infty$ (3)

Question 6 continued

The same construction as described above is applied externally to the three sides of an equilateral triangle of side length a .

Given that the limit of the area of the resulting shape is $26\sqrt{3}$

(g) find the value of a .

(3)

(+S2)

(Total for Question 6 is 24 marks)

7. A circle C has centre $X(a, b)$ and radius r .

A line l has equation $y = mx + c$

(a) Show that the x coordinates of the points where C and l intersect satisfy

$$(m^2 + 1)x^2 - 2(a - m(c - b))x + a^2 + (c - b)^2 - r^2 = 0 \quad (2)$$

Given that l is a tangent to C ,

(b) show that

$$c = b - ma \pm r\sqrt{m^2 + 1} \quad (6)$$

The circle C_1 has equation

$$x^2 + y^2 - 16 = 0$$

and the circle C_2 has equation

$$x^2 + y^2 - 20x - 10y + 89 = 0$$

(c) Find the equations of any lines that are normal to both C_1 and C_2 , justifying your answer.

(3)

(d) Find the equations of all lines that are a tangent to both C_1 and C_2

[You may find the following Pythagorean triple helpful in this part: $7^2 + 24^2 = 25^2$]

(11)

(+S2)

(Total for Question 7 is 24 marks)

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